

SET A (w13)

CSE427 Quiz 3
Summer 2026
Duration: 25 Minutes

Name:

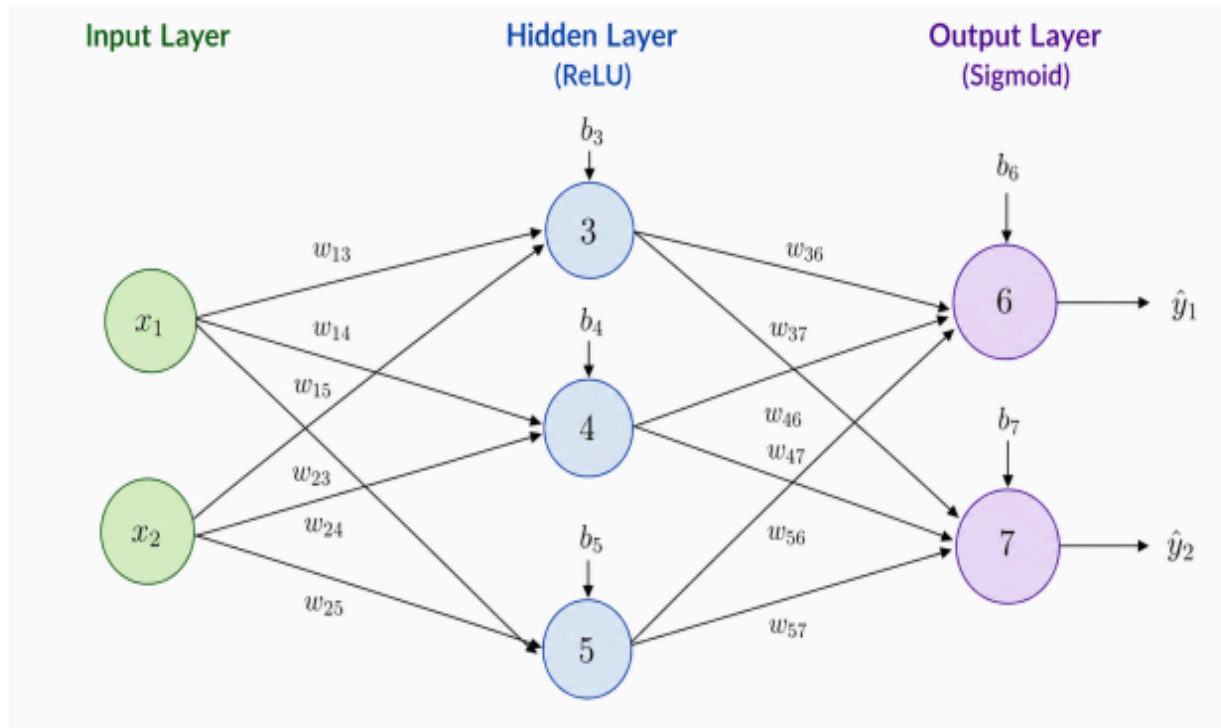
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Question [15 Points]

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{13}}$.

SET B (w14)

CSE427 Quiz 3
Summer 2026
Duration: 25 Minutes

Name:

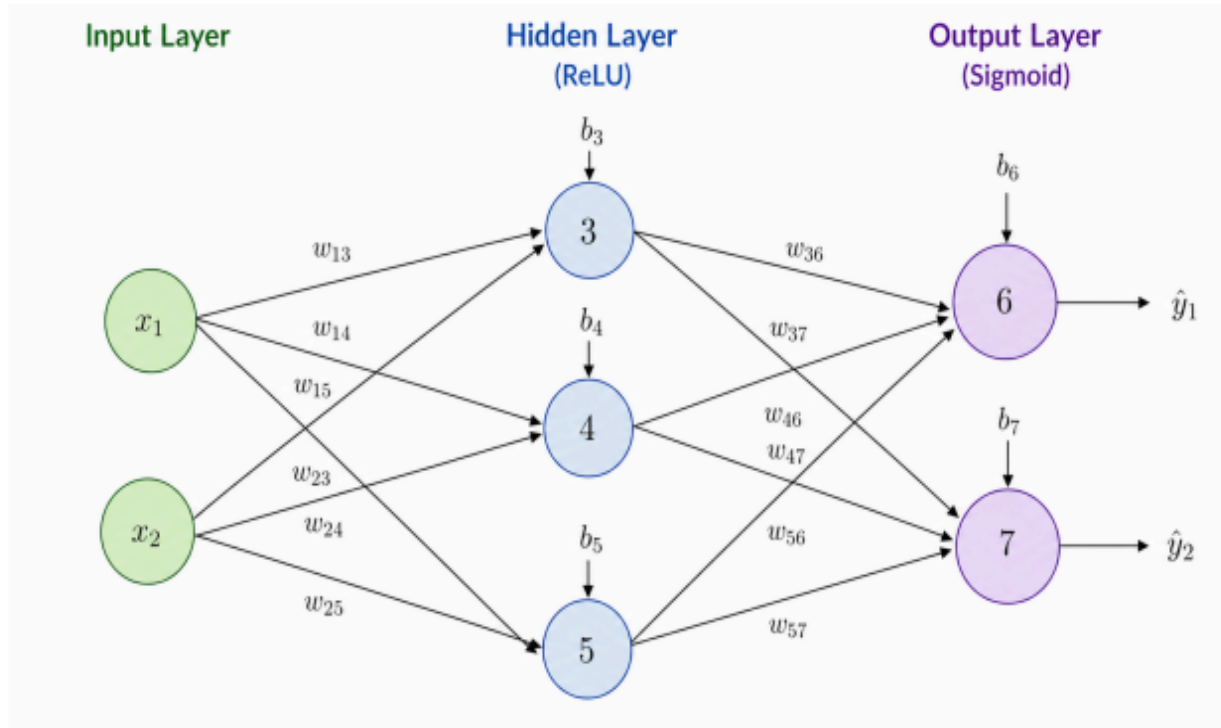
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Question [15 Points]

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{14}}$.

SET C (w15)

CSE427 Quiz 3
Summer 2026
Duration: 25 Minutes

Name:

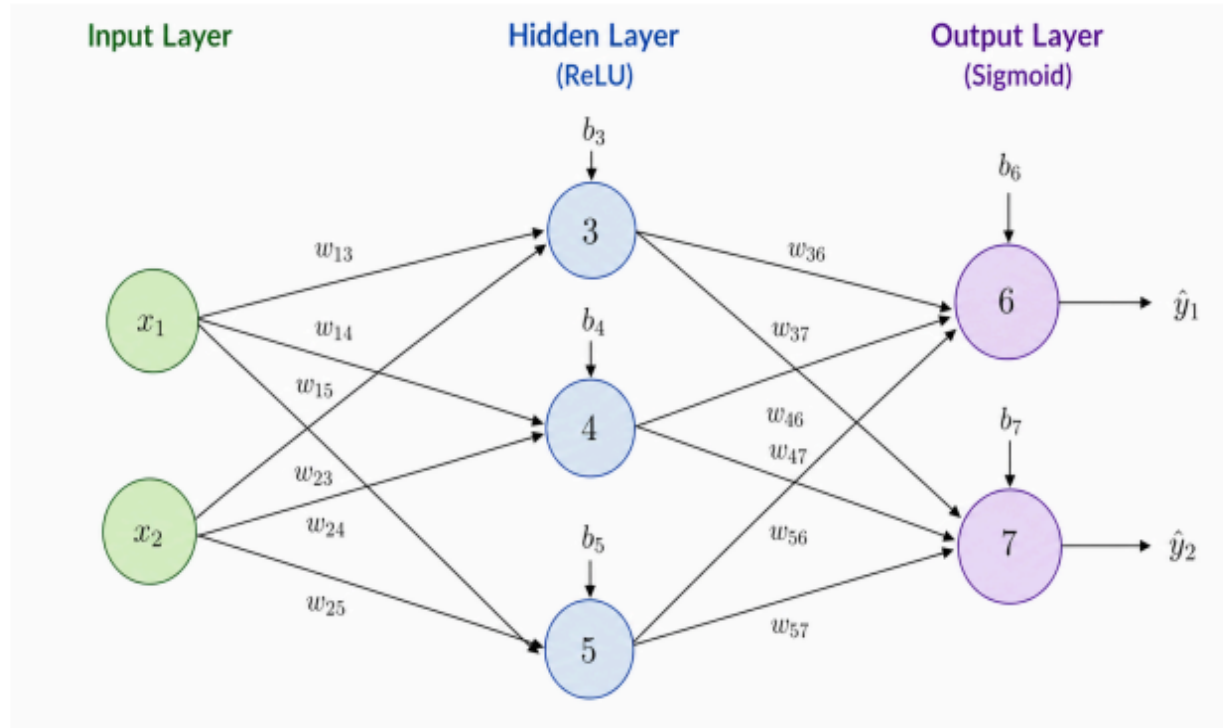
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Question [15 Points]

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{15}}$.

SET D (w23)

CSE427 Quiz 3
Summer 2026
Duration: 25 Minutes

Name:

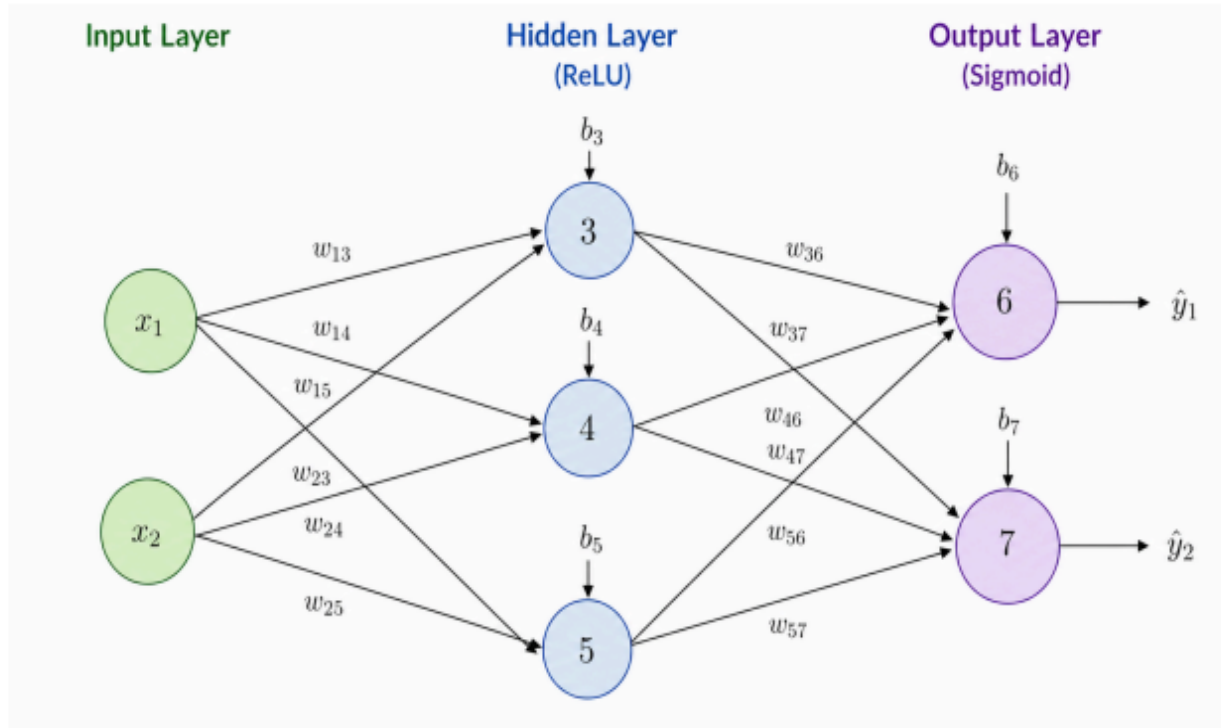
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Question [15 Points]

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{23}}$.

SET E (w24)

CSE427 Quiz 3
Summer 2026
Duration: 25 Minutes

Name:

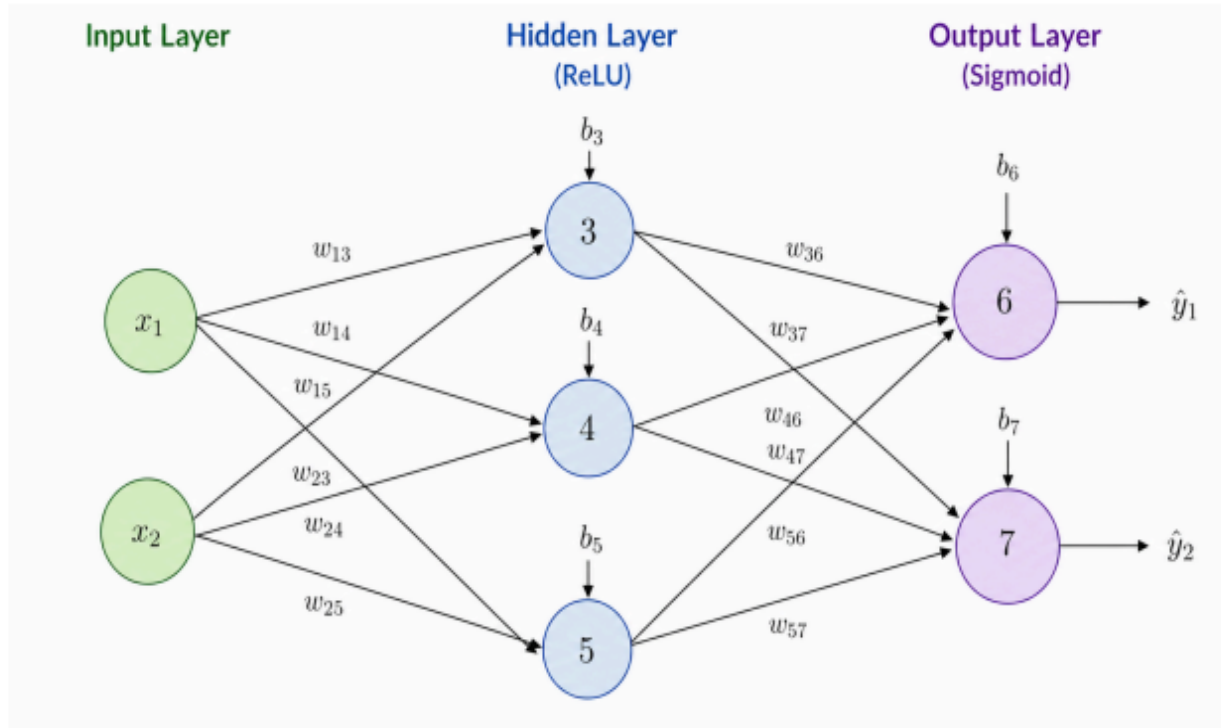
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Question [15 Points]

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{24}}$.

SET F (w25)

CSE427 Quiz 3
Summer 2026
Duration: 25 Minutes

Name:

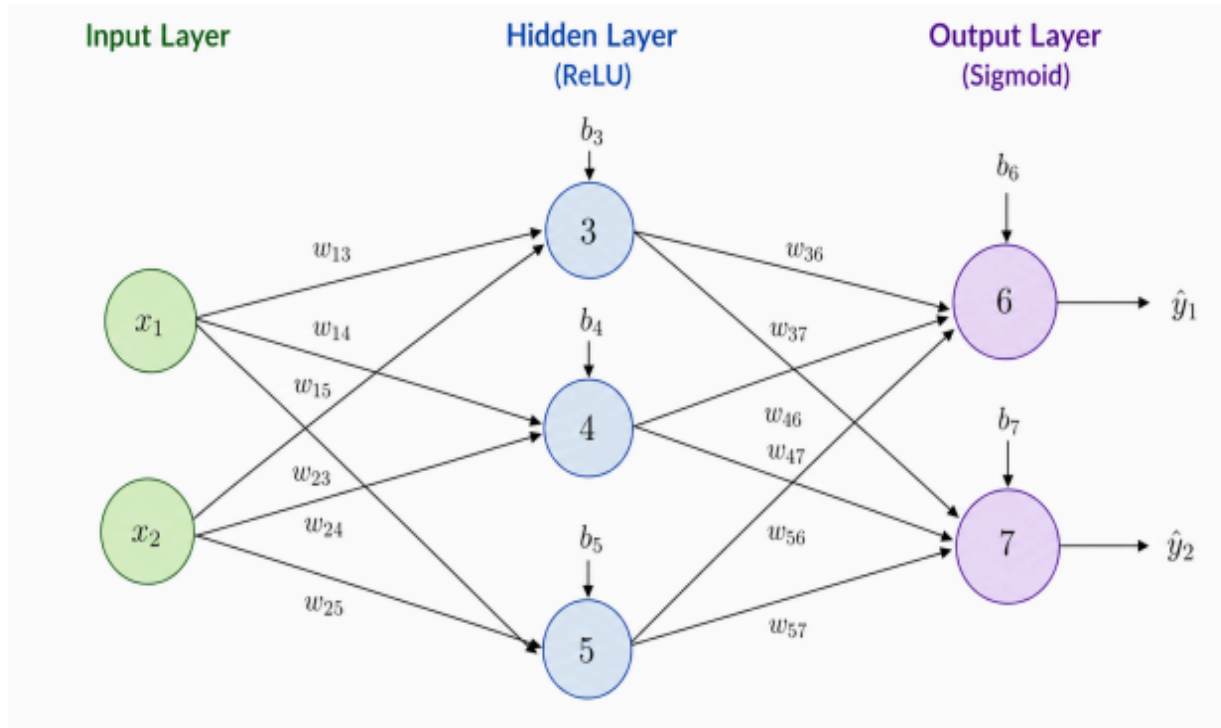
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Question [15 Points]

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

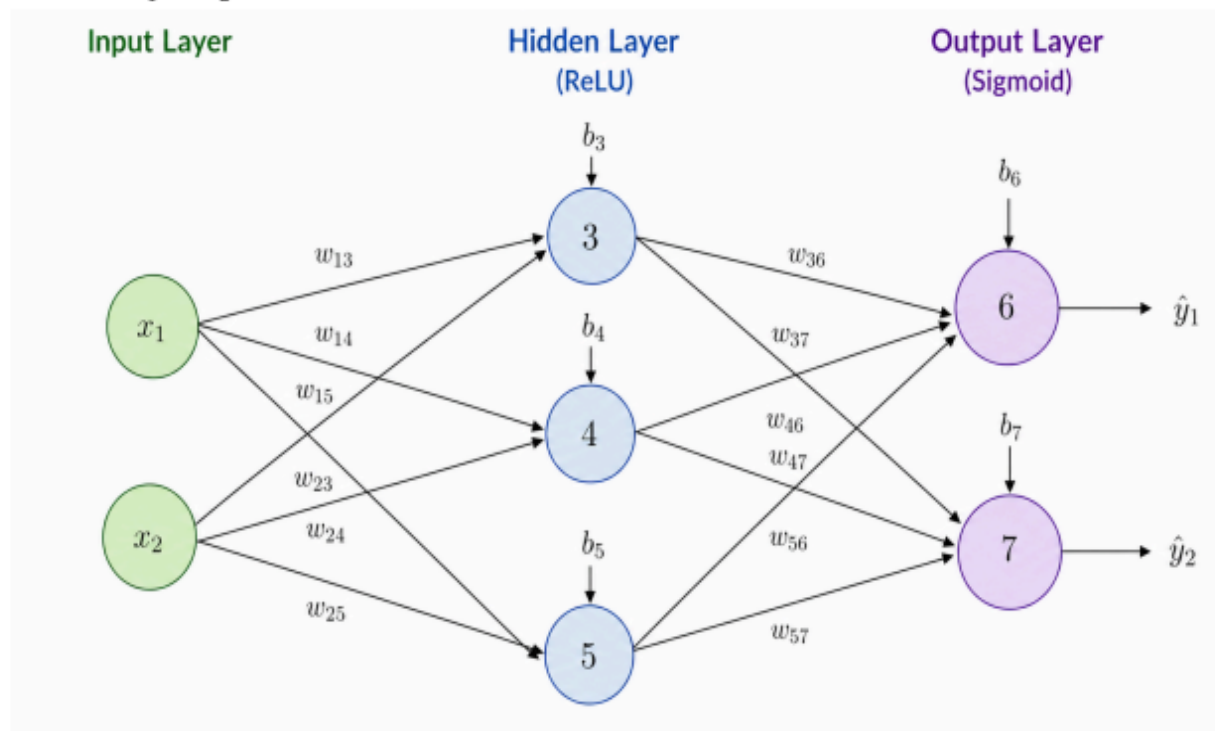
Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{25}}$.

Solution

Solution

SET A

Consider the following neural network with two inputs (x_1 & x_2) and two outputs (y_1 & y_2). The only hidden layer consists of three neurons with ReLU activation function, and the output layer consists of two neurons (that predict $\hat{y}_1$ & $\hat{y}_2$) with Sigmoid activation function.



Suppose for a specific input sample, the target output is: ($y_1 = 1$ & $y_2 = 0$).

Now, consider the following Loss Function:

$$Loss = -[\ln(\hat{y}_1) + \ln(1 - \hat{y}_2)]$$

Perform the step-by-step derivation for $\frac{\partial Loss}{\partial w_{25}}$.

Solution:

Forward Pass: (Same for All Sets)

Hidden-layer pre-activation values and ReLU activations:

$$\begin{aligned}z_3 &= x_1 w_{13} + x_2 w_{23} + b_3 ; a_3 = \text{ReLU}(z_3) \\z_4 &= x_1 w_{14} + x_2 w_{24} + b_4 ; a_4 = \text{ReLU}(z_4) \\z_5 &= x_1 w_{15} + x_2 w_{25} + b_5 ; a_5 = \text{ReLU}(z_5)\end{aligned}$$

Output-layer pre-activation values and Sigmoid activations:

$$\begin{aligned}z_6 &= a_3 w_{36} + a_4 w_{46} + a_5 w_{56} + b_6 ; \hat{y}_1 = \sigma(z_6) \\z_7 &= a_3 w_{37} + a_4 w_{47} + a_5 w_{57} + b_7 ; \hat{y}_2 = \sigma(z_7)\end{aligned}$$

Backward Pass: (Using Delta Notation) (Same for All Sets)

For sigmoid output with BCE loss, the output-layer error terms are:

$$\begin{aligned}\delta_6 &= \partial L / \partial z_6 = (\partial L / \partial \hat{y}_1) \cdot (\partial \hat{y}_1 / \partial z_6) = -(1/\hat{y}_1) \cdot [\hat{y}_1(1-\hat{y}_1)] = \hat{y}_1 - 1 \\ \delta_7 &= \partial L / \partial z_7 = (\partial L / \partial \hat{y}_2) \cdot (\partial \hat{y}_2 / \partial z_7) = -[-1/(1-\hat{y}_2)] \cdot [\hat{y}_2(1-\hat{y}_2)] = \hat{y}_2\end{aligned}$$

The hidden-layer error terms are obtained by backpropagating the output errors through the outgoing weights:

$$\begin{aligned}\delta_3 &= \text{ReLU}'(z_3)[w_{36} \delta_6 + w_{37} \delta_7] \\ \delta_4 &= \text{ReLU}'(z_4)[w_{46} \delta_6 + w_{47} \delta_7] \\ \delta_5 &= \text{ReLU}'(z_5)[w_{56} \delta_6 + w_{57} \delta_7]\end{aligned}$$

Substituting $\delta_6 = \hat{y}_1 - 1$ and $\delta_7 = \hat{y}_2$ gives:

$$\begin{aligned}\delta_3 &= \text{ReLU}'(z_3)[w_{36}(\hat{y}_1 - 1) + w_{37}\hat{y}_2] \\ \delta_4 &= \text{ReLU}'(z_4)[w_{46}(\hat{y}_1 - 1) + w_{47}\hat{y}_2] \\ \delta_5 &= \text{ReLU}'(z_5)[w_{56}(\hat{y}_1 - 1) + w_{57}\hat{y}_2]\end{aligned}$$

Gradient Calculation: (Using Delta Notation)

The weight w_{13} connects input x_1 to hidden neuron 3. Therefore, we use the hidden-layer error term δ_3 .

$$\begin{aligned}\partial L / \partial w_{13} &= \delta_3 (\partial z_3 / \partial w_{13}) \\ \partial z_3 / \partial w_{13} &= x_1 \\ \partial L / \partial w_{13} &= x_1 \delta_3 \\ \partial L / \partial w_{13} &= x_1 \text{ReLU}'(z_3)[w_{36}(\hat{y}_1 - 1) + w_{37}\hat{y}_2]\end{aligned}$$

[Alternatively, without Delta Notation] Backward Pass & Gradient Calculation:

The weight w_{13} affects both output neurons through hidden activation a_3 . Hence, the two chain-rule paths are added.

$$\begin{aligned}\partial L / \partial w_{13} &= (\partial L / \partial z_6) (\partial z_6 / \partial a_3) (\partial a_3 / \partial z_3) (\partial z_3 / \partial w_{13}) \\ &+ (\partial L / \partial z_7) (\partial z_7 / \partial a_3) (\partial a_3 / \partial z_3) (\partial z_3 / \partial w_{13})\end{aligned}$$

Substitute the local derivatives:

$$\begin{aligned}\partial L / \partial z_6 &= (\partial L / \partial \hat{y}_1) \cdot (\partial \hat{y}_1 / \partial z_6) = -(1 / \hat{y}_1) \cdot [\hat{y}_1 (1 - \hat{y}_1)] = \hat{y}_1 - 1 \\ \partial L / \partial z_7 &= (\partial L / \partial \hat{y}_2) \cdot (\partial \hat{y}_2 / \partial z_7) = -[1 / (1 - \hat{y}_2)] \cdot [\hat{y}_2 (1 - \hat{y}_2)] = \hat{y}_2 \\ \partial z_6 / \partial a_3 &= w_{36} \\ \partial z_7 / \partial a_3 &= w_{37} \\ \partial a_3 / \partial z_3 &= \text{ReLU}'(z_3) \\ \partial z_3 / \partial w_{13} &= x_1\end{aligned}$$

Therefore:

$$\begin{aligned}\partial L / \partial w_{13} &= (\hat{y}_1 - 1) w_{36} \text{ReLU}'(z_3) x_1 + \hat{y}_2 w_{37} \text{ReLU}'(z_3) x_1 \\ \partial L / \partial w_{13} &= x_1 \text{ReLU}'(z_3) [w_{36}(\hat{y}_1 - 1) + w_{37} \hat{y}_2]\end{aligned}$$

Summarized Solution of All Sets

Set	Gradient	where, $z_3 = x_1 w_{13} + x_2 w_{23} + b_3$ $z_4 = x_1 w_{14} + x_2 w_{24} + b_4$ $z_5 = x_1 w_{15} + x_2 w_{25} + b_5$
A	$\partial L / \partial w_{13}$ $= x_1 \delta_3$ $= x_1 \text{ReLU}'(z_3) [w_{36}(\hat{y}_1 - 1) + w_{37} \hat{y}_2]$	
B	$\partial L / \partial w_{14}$ $= x_1 \delta_4$ $= x_1 \text{ReLU}'(z_4) [w_{46}(\hat{y}_1 - 1) + w_{47} \hat{y}_2]$	
C	$\partial L / \partial w_{15}$ $= x_1 \delta_5$ $= x_1 \text{ReLU}'(z_5) [w_{56}(\hat{y}_1 - 1) + w_{57} \hat{y}_2]$	
D	$\partial L / \partial w_{23}$ $= x_2 \delta_3$ $= x_2 \text{ReLU}'(z_3) [w_{36}(\hat{y}_1 - 1) + w_{37} \hat{y}_2]$	
E	$\partial L / \partial w_{24}$ $= x_2 \delta_4$ $= x_2 \text{ReLU}'(z_4) [w_{46}(\hat{y}_1 - 1) + w_{47} \hat{y}_2]$	
F	$\partial L / \partial w_{25}$ $= x_2 \delta_5$ $= x_2 \text{ReLU}'(z_5) [w_{56}(\hat{y}_1 - 1) + w_{57} \hat{y}_2]$	